

1 Correlation

1. Toy dataset: Correlation

When analyzing data, a first important aspect to understand it is to analyze the relation between variables. For instance, if I want to predict if chocolate can cure cancer, I can start by checking if people eating more chocolate tend to have less cancer. We can use correlation coefficients to do so naively. However, we will see that these coefficients should be analyzed with caution

- (a) Load the dataset `coffee_effects.csv` found on the class website
- (b) Plot the first few lines to check the content (`.head(2)`), or simply write the name of the dataframe in a notebook)
- (c) Check the Pearson correlation between variables. You can directly use `df.corr()` for instance. You can plot the correlation matrix for instance with `seaborn` library with a command such as `sns.heatmap(correlation, annot=True, cmap='coolwarm', fmt=".2f")`
- (d) Do the same with Spearman (option `method='spearman'`)
- (e) Observe the correlation values and try to interpret the relation between the variables
- (f) Draw a scatterplot with the caffeine consumption on the x-axis, and another variable in the y-axis. Repeat for all 3 variables. Use `plotly` library to have an interactive plot, for instance with the command `px.scatter(df, x='caffeine_consumption_mg', y='productivity')`
- (g) For each relation, you should observe something unusual. Ask yourself questions such as: is the relation linear? Is it monotonous? How strong is the relation between changes in the two variables?
- (h) Using this information, can you propose a strategy to improve the productivity of a company? Should it offers free coffee to employees ? If yes, unlimited ?

2. Mastering the scores

To be sure to understand correctly what is going on, we will write code to compute manually simple scores. For each question, use the native function in python corresponding to those scores to check that your function is correct

- (a) Write a function to compute the variance of a variable
- (b) Write a function to compute the covariance between two variables
- (c) Using the covariance, write a function to compute the Pearson (linear) correlation coefficient
- (d) Using the Pearson CC function, write a function that compute the Spearman CC (you can use for instance `scipy.stats.rankdata`).